- *Sensor fusion and ISR processing*: Software to geo-register sensor feeds, perform real-time analytics (object detection, change detection), and generate intelligence products.
- *Network and cybersecurity layer*: Encryption, anti-spoofing, resilient routing, and intrusion detection to safeguard command and data links.
- *Health management and FDIR*: Continuous monitoring of avionics, propulsion, and structural health; automatic corrective actions and contingency handling.
- *Human–machine interface (HMI)*: Ground control station software for mission planning, real-time visualization, collaborative tasking, and rules-of-engagement enforcement.

### 4.3.4 System integration and functional operation

Third-generation UAVs are engineered as cohesive systems where hardware and software interplay across closed control loops:

1. *Pre-mission*: Mission plan and sensor prescriptions are uploaded via the GCS; calibration and system checks execute automatically.
2. *Launch and transit*: Autopilot and navigation modules execute takeoff and place the UAV on optimized transit routes using GNSS/INS and satellite link monitoring.
3. *On-station operations*: Sensors activate, and the mission computer autonomously manages sensor employment (e.g., cued EO/IR slewing, SAR strip mapping), while onboard processors run ATR and compress data for transmission.
4. *Dynamic re-tasking*: Operators or higher-level C2 nodes can retask the platform in real time via resilient datalink; onboard autonomy can accept high-level objectives and replan locally to satisfy mission constraints.